

◆ LEVEL 1 — READING

1. Display all data

```
SELECT * FROM superstore;
```

Insight :

The dataset contains 21 columns and 9994 rows, providing a comprehensive view of sales transactions, including revenue, profit, discounts, and customer information.

2. Display only: Sales, Profit, Category

```
SELECT Sales, Profit, Category FROM superstore;
```

Insight :

Sales and profit vary significantly across categories. Some transactions generate high revenue but negative profit, indicating that revenue alone is not a reliable measure of performance.

3. Display the first 15 rows

```
SELECT * FROM superstore  
LIMIT 15;
```

Insight :

The first rows confirm the structure and consistency of the dataset, allowing quick validation of column types and values before deeper analysis.

4. Sort sales from highest to lowest

```
SELECT * FROM superstore  
ORDER BY Sales DESC;
```

Insight :

A small number of transactions generate extremely high sales (up to 22,638), suggesting the presence of high-value orders that may heavily influence overall revenue.

◆ LEVEL 2 — SIMPLE FILTERING

5. Show sales greater than 500

```
SELECT * FROM superstore  
WHERE Sales > 500;
```

Insight :

High-value transactions (>500) represent a significant portion of the dataset, indicating that large orders are relatively common.

6. Show orders with negative profit

```
SELECT * FROM superstore  
WHERE Profit < 0;
```

Insight :

Loss-making transactions range from small losses to very large ones, highlighting potential issues in pricing or discount strategies.

7. Show orders from the "West" region

```
SELECT * FROM superstore  
WHERE Region = 'West';
```

Insight :

The West region contains a large number of transactions, suggesting it is one of the most active and strategically important markets.

8. Show sales between 100 and 300

```
SELECT * FROM superstore  
WHERE Sales BETWEEN 100 AND 300;
```

Insight :

A large number of transactions fall within the mid-range sales segment, indicating that most orders are moderate in value rather than high-value.

◆ LEVEL 3 — LOGICAL CONDITIONS

9. Sales > 500 AND positive profit

```
SELECT * FROM superstore  
WHERE Sales > 500 AND Profit > 0;
```

Insight :

Not all high-value sales are profitable, showing that revenue does not guarantee financial success.

10. Region "East" OR "West"

```
SELECT * FROM superstore  
WHERE Region IN ('East', 'West');
```

Insight :

The majority of transactions occur in East and West regions, indicating that these areas dominate business activity.

11. Exclude the "Furniture" category

```
SELECT * FROM superstore  
WHERE Category != 'Furniture';
```

Insight :

Non-furniture categories represent a large portion of transactions, suggesting diversification across product types.

12. Negative profit AND discount > 0.2

```
SELECT * FROM superstore  
WHERE Profit < 0 AND Discount > 0.2;
```

Insight :

A large number of loss-making transactions are associated with high discounts (>20%), suggesting that aggressive discounting negatively impacts profitability.

◆ LEVEL 4 — BUSINESS ANALYSIS

13. High sales BUT negative profit

```
SELECT * FROM superstore  
WHERE Sales > 500 AND Profit < 0;
```

Insight :

Many high-value transactions still generate losses, indicating that large sales are often driven by excessive discounts or high costs.

14. Low sales BUT positive profit

```
SELECT * FROM superstore  
WHERE Sales < 100 AND Profit > 0;
```

Insight :

A large number of low-value transactions are profitable, suggesting that smaller sales may be more efficient and less risky.

15. Highest discounts

```
SELECT * FROM superstore  
ORDER BY Discount DESC;
```

Insight :

Discounts can reach up to 80%, which is extremely high and likely unsustainable, potentially explaining the presence of significant losses.

16. Most profitable orders

```
SELECT * FROM superstore  
ORDER BY Profit DESC;
```

Insight :

Some transactions generate very high profits, indicating strong revenue opportunities when pricing and costs are well managed.

17. Least profitable orders

```
SELECT * FROM superstore  
ORDER BY Profit ASC;
```

Insight :

Certain transactions generate severe losses, highlighting critical inefficiencies that need to be addressed.

◆ LEVEL 5 — SIMPLE CALCULATIONS

18. Profit ratio

```
SELECT *,  
Profit / Sales AS profit_ratio  
FROM superstore;
```

Insight :

Profitability varies significantly across transactions, with both strong gains and negative returns.

19. Negative profit ratio

```
SELECT *,  
Profit / Sales AS profit_ratio  
FROM superstore  
WHERE Profit / Sales < 0;
```

Insight :

A significant number of transactions have negative profitability, confirming that many sales are actually generating losses.

20. High profit ratio (> 0.3)

```
SELECT *,  
Profit / Sales AS profit_ratio  
FROM superstore  
WHERE Profit / Sales > 0.3;
```

Insight :

Many transactions have high profit ratios, indicating strong profitability potential, but this contrasts with the presence of many loss-making transactions, showing inconsistency in performance.